

A DOCUMENTATION FOR THE PROJECT IN Programming 2

BSIT 1-1

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PROJECT OVERVIEW

The HR Applicant Process Windows Application is a desktop-based recruitment management system developed using C# language with Windows Forms and a relational database in use of Microsoft Access. The system is designed to streamline the hiring process by supporting two main user types which are the Applicants and HR Personnel.

Applicants can register an account, complete their profile, browse job vacancies, submit applications, upload required documents, and track their application status in real time. On the other hand, HR Staff and HR Managers can review submissions, screen applicants, schedule and evaluate interviews, and make final hiring decisions.

The system enforces role-based access control to ensure that each user only performs actions within their designated responsibilities, maintaining data integrity and process accountability throughout the entire recruitment workflow.

SYSTEM ARCHITECTURE

The following tech stack are used for the creation of the project:

1. C# Language
2. Microsoft Access file
3. GitHub Repository
4. Integrated Development Environment (IDE)

DATABASE DESIGN

The system uses Microsoft Access as its database management system, chosen for its lightweight setup and compatibility with C# Windows Forms applications through OLE DB connection. The database contains all the fifteen (15) necessary tables to support the full recruitment workflow, including user accounts, job vacancies, applications, documents, screening results, interview schedules, and hiring decisions.

Each table is structured with defined primary keys and relationships to maintain data integrity and avoid redundancy. Referential integrity is enforced through table relationships within the Access database to ensure that linked records remain consistent across all modules.

(Sample screenshot of the database).

This screenshot shows the Microsoft Access interface with the 'JobVacancies' table selected. The table has five columns: JobID, JobTitle, JobDetail, JobSlots, and Status. The data is as follows:

JobID	JobTitle	JobDetail	JobSlots	Status
1	IT Support	Ste Troubleshoot	3	Open
2	Junior Program	Develop and n	2	Open
3	Database Assi	Maintain recor	1	Open
4	HR Assistant	Assist in recru	2	Open
5	Administrative	Handle docum	4	Open
*	(New)		0	

This screenshot shows the Microsoft Access interface with the 'Users' table selected. The table has four columns: UserID, Username, Password, and RoleID. The data is as follows:

UserID	Username	Password	RoleID
1	manager	1234	2
2	john	op	1
3	admin	4321	2
4	ace	6767	1
*	(New)		0

SYSTEM FLOW

Applicant Flow

The process begins when the applicant launches the system and is presented with the login page. If the applicant does not have an account, they may proceed to the registration page to create one. Once registered, the applicant logs in and is directed to the dashboard. From there, the applicant can complete their profile, browse available job vacancies, and submit an application for a chosen position. After submission, the applicant may upload the required documents and monitor their application status through the status tracking page.

HR Flow

The HR process begins when an HR user logs in using their assigned credentials. Based on their role, they are granted access to specific features of the system. HR Staff can search and filter applicants, review submitted applications, lock applications for review, and conduct screening to determine if an applicant is qualified. Qualified applicants are then scheduled for an interview, and results are recorded through the interview evaluation module. Once evaluated, the application is forwarded for final review. Only the HR Manager or Admin can issue the final hiring decision, marking the applicant as Accepted, Rejected, or On Hold. Every status change is automatically recorded in the application status history.

AI USAGE

Artificial Intelligence tools were used in this project solely as a learning aid and guidance resource. Specifically, AI was consulted to help the team better understand the overall recruitment workflow, clarify the sequence of application status changes, and gain a clearer picture of how each module connects to one another.

AI was also used to assist in explaining concepts such as role-based access control, database table relationships, and form validation logic, serving as a supplementary reference similar to reading tutorials or documentation. All actual coding, design decisions, database construction, and implementation were performed by the team members themselves.

CONCLUSION

Building the HR Applicant Process Windows Application gave our group a better understanding of how to develop a real-world desktop system using C# Windows Forms and Microsoft Access. Through this project, we were able to apply what we learned in class into an actual working application that handles recruitment processes from application submission up to the final hiring decision.

Our group also learned the importance of teamwork, planning, and communication in completing a system with multiple modules and features. Each part of the project helped us grow not just as programmers, but also as problem solvers.

Overall, this capstone project was a meaningful experience that prepared our group for bigger challenges in our future careers in software development.

Some photo took during the implementation :

```
=== Applicant Dashboard ===
1. View Profile
2. Update Profile
3. Browse Job Vacancies
4. View My Applications
5. Logout
Choose an option: 1

=== Your Profile ===
Name: John Code
Contact: 2021
Address: Dyan
Education: Alberts
Skills: Codings
```

```
HRAndApplicantSystem
1 using System;
2 using HRApplicantSystem.Database;
3
4 class Program
5 {
6     static void Main()
7     {
8         try
9         {
10             using (var conn = DatabaseHelper.GetConnection())
11             {
12                 conn.Open();
13                 Console.WriteLine("Database connected successfully!");
14             }
15         }
16         catch (Exception ex)
17         {
18             Console.WriteLine("Connection failed: " + ex.Message);
19         }
20     }
21 }
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